

FSTS Website Operating System™ — Full Project Status Report

Generated: July 16, 2026
Prepared for: New Dedicated Dashboard Project

Project Separation Summary

This report captures the complete task history from the original combined repository before the official project separation.

Dashboard Project	FSTS Website Operating System™ (WOS™)
GitHub	https://github.com/thefsts/FSTS-client-Dashboard-for-sites-
Vercel	https://vercel.com/fullstacksolutions/fsts-client-dashboard-for-sites-api-server
Corsair Website	https://github.com/thefsts/Corsair-Tactical-Solutions
Vercel	https://vercel.com/fullstacksolutions/corsair-tactical-solutions

Platform Stack

Layer	Technology
Monorepo	pnpm workspaces
Runtime	Node.js 24
Language	TypeScript 5.9
Frontend	React + Vite
Backend API	Express 5
Realtime/Serverless	Convex
Database	PostgreSQL + Drizzle ORM
Auth	Clerk (multi-tenant)
Validation	Zod v4 + drizzle-zod
API Codegen	Orval (from OpenAPI spec)
Build	esbuild (CJS bundle)
Payments	Square Commerce
Deployment	Vercel

Task Status Overview

Status	Count
Merged (Complete)	63
Draft (Pending)	26
Cancelled	7
Total	**96**

COMPLETED WORK (63 Tasks Merged)

Corsair Website — Content & Courses

#	Task
#1	Push course updates to GitHub
#2	Add real photos to the 5 new course pages
#3	Show the new courses on the homepage carousel
#4	Translate new course names into all 9 supported languages
#5	Apply the new courses patch so all 5 course pages go live on the website
#6	Replace the security officer uniform photo with a real Corsair instructor photo
#7	Translate the 3 new homepage course cards into all 9 languages
#8	Let the homepage show all 9 courses instead of just 6 or 9 fixed slots
#9	Translate remaining English-only offering labels into all 9 languages
#10	Add Non-Lethal Defense and Firearm Re-Qualification to the homepage course carousel
#11	Translate the 5 new course detail pages into all 9 languages
#12	Replace all remaining stock/placeholder photos with real Corsair training photos
#14	Compress large training photos so the site loads faster

Corsair Website — Course Navigation & Design System

#	Task
#24	Let each course card show a level badge (Beginner, Intermediate, Advanced)
#25	Make homepage course cards wrap into a more readable grid

#26	Add a 'View All Courses' grid page showing every course at a glance
#28	Add level-filter tabs to the All Courses page so visitors can browse by skill level
#29	Move course data into a shared file so both pages stay in sync automatically
#30	Keep course card layout consistent between homepage and All Courses page
#33	Apply card design updates everywhere at once by extracting shared styles into CourseCard
#34	Apply the named-constants pattern to other shared UI components
#35	Apply the named-constants pattern to the ui/ component library files
#37	Extract shared design tokens into a central constants file

Corsair Website — Square Commerce

#	Task
#15	Square Sandbox + catalog integration
#22	Save bookings and email confirmations after payment
#23	Itemize receipts by routing checkout through Square Orders
#70	Phase 5 — Square Commerce (full integration)

Corsair Website — Visual Regression Testing

#	Task
#36	Add automated visual regression tests to catch style regressions
#38	Keep visual tests green when new mockup pages are added
#39	Run visual regression tests automatically on every code change
#42	Send a visual diff report when style changes are detected
#43	Warn when a new mockup page has no saved visual baseline yet
#44	Auto-generate missing baselines on first run instead of failing

FSTS-WOS Dashboard — Backend & Infrastructure

#	Task
#49	Migrate dashboard backend to Convex
#50	Push Convex backend functions to production so the dashboard has live data
#51	Remove the old Express API server and unused library packages
#52	Confirm login and first-time data sync work correctly end-to-end
#53	Remove leftover build artifacts so the repo stays clean
#55	Store the Convex deploy key as a secret so backend updates don't need manual key entry
#57	Confirm a Convex backend change deploys end-to-end without errors

FSTS-WOS Dashboard — Authentication & Security

#	Task
#56	Prevent a deactivated user from silently accessing any dashboard page
#61	Fix Vercel 404 on dashboard routes (rewrite rule + missing env vars)
#62	Show a clear error message when the wrong Clerk key type is used in production
#63	Show a clear error when VITE_CONVEX_URL is missing in production
#64	Catch wrong Clerk key before it crashes the deployed app (automated check)

FSTS-WOS Dashboard — Core Platform Phases

#	Task
#67	Phase 1 — CMS Pipelines (wire static data to Convex)
#68	Phase 2 — Missing Dashboard Modules
#69	Phase 3 — Drag-and-Drop Form Builder
#78	Dashboard Experience Upgrade (Live Preview, Smart Images, AI Assistant, Health Center)
#83	WOS Phase 1 — Payment Connector Framework
#84	WOS Phase 2 — Website Settings
#85	WOS Phase 3 — Global Design Lock
#86	WOS Phase 9 — Client Permissions
#87	WOS Phase 11 — Operon CRM Connector
#88	WOS Phase 8 — Automation Engine
#89	WOS Phase 10 — Agency Edition
#90	WOS Phase 12 — Final Platform Standard
#91	Let other editor pages use Live Preview and Publish Validation
#92	Make health scan categories clickable — link to the relevant dashboard section
#93	Fix pre-existing TypeScript errors blocking a clean build
#94	Let the AI Assistant answer questions about a specific page's real content

FSTS-WOS Dashboard — Product Architecture

#	Task
#95	Define permanent product boundaries between FSTS-WOS and Operon CRM (incl. Website Reviews Module)
#96	Enforce product boundaries in code review so CRM features can't silently land in FSTS-WOS

OPEN DRAFT TASKS (26 Pending)

Corsair Website — Remaining Work

#	Task	Depends On
#13	Translate remaining English-only course card text (Itc, firstShots, defensiveShooting, basicHandgun, level3Security) into all 9 languages	#7
#16	Convert training photos to WebP for faster loading on all devices	#14
#17	Add lazy-loading to training photos so the page feels instant on first visit	#14
#18	Translate remaining English-only homepage section text into all 9 languages	#9
#19	Show all 11 courses in the homepage carousel (not just the first 9)	#10
#20	Add detail pages for Non-Lethal Defense and Firearm Re-Qualification	#10
#21	Add native-language detail content to the 5 new course pages	#11
#27	Wire level badge translations into the live homepage course cards	#24
#31	Remember the selected level filter when visitors navigate back to All Courses	#28
#32	Translate level-filter tab labels into all 9 languages on the All Courses page	#28
#40	Audit remaining hardcoded Tailwind strings outside COMPONENT_CLASSES objects	#37
#41	Add tokens for typography and spacing scales used across cards, dialogs, and lists	#37
#45	Share the visual diff report as a Slack message when style changes are detected	#42
#46	Show a live preview of the visual diff report inside the app	#42
#74	Surface Square Commerce data to the Corsair website (catalog, checkout, order confirmation)	—
#77	Connect the new Phase 2 public endpoints to the Corsair website	—

FSTS-WOS Dashboard — Remaining Work

#	Task	Depends On
#54	Keep Convex deploy key in sync so backend pushes don't break silently	—
#58	Confirm a deactivated account cannot access any dashboard page, even after bypassing the sign-in screen	—
#59	Show a clear Deactivated badge on the user row so admins know the account is already blocked before editing	—
#66	Prevent a silent crash when VITE_CONVEX_URL points to the wrong deployment	—
#75	Let site owners control which modules show in their sidebar from the admin panel	—
#76	Prevent broken webhook deliveries when Square signature key is not configured	—
#79	Show form submissions inside Contact Inbox so nothing gets missed	—
#80	Let site visitors fill out forms embedded directly in site pages	—
#81	Prevent form submissions from being lost if Convex is briefly unavailable	—
#82	Catch broken form builder layouts before they ship to site admins	—

CANCELLED TASKS (7)

#	Task	Reason
#47	Commit auto-generated baselines so CI starts clean for every developer	Superseded
#48	Run visual regression tests automatically when mockup files change	Superseded
#60	Catch broken Convex deploys before they block other tasks	Superseded
#65	Catch a missing or wrong CONVEX_DEPLOY_KEY before a backend push silently fails	Superseded
#71	Link courses directly to Square catalog items from the Course editor	Cancelled
#72	Attach a Square item to paid events so registration fees sync automatically	Cancelled
#73	Prevent a Square webhook event from writing to the wrong site's payment record	Cancelled

FSTS-WOS Dashboard — Completed Module Inventory

The following platform modules were built and merged. All are part of the Dashboard product moving to the new project:

- Convex Backend — Serverless functions, real-time data, production deployed
- Clerk Authentication — Multi-tenant auth with RBAC, deactivation enforcement
- CMS Pipelines — Static data wired to Convex (Phase 1)
- Missing Dashboard Modules — Phase 2 gaps filled
- Drag-and-Drop Form Builder — Phase 3
- Payment Connector Framework — Phase 1 (Square, extensible to other providers)
- Website Settings — Phase 2
- Global Design Lock — Phase 3 (visual regression guard, 69 automated tests)
- Automation Engine — Phase 8
- Client Permissions — Phase 9
- Agency Edition — Phase 10
- Operon CRM Connector — Phase 11 (provider-agnostic schema, Operon is provider #1)
- Final Platform Standard — Phase 12
- Live Preview — Real-time page preview in editor
- Publish Validation — Pre-publish checks
- Smart Image Manager — Image handling
- AI Dashboard Assistant — Page-aware AI Q&A (scope-limited to WOS only)
- Website Health Command Center — Health scan with clickable categories
- Product Boundary Enforcement — Documented in docs/product-boundaries.md + automated code review check

Key Architecture Decisions to Carry Forward

- Product boundary is law. FSTS-WOS owns website operations. Operon CRM owns customer/lead management. The only connection between them is the Operon Connector. See docs/product-boundaries.md.
- Convex is the sole backend. The old Express API server was removed. All data mutations and queries go through Convex functions.
- Clerk handles all auth. Multi-tenant. First-ever signup bootstraps the superadmin. Deactivated users are blocked at the Convex middleware layer, not just the UI.
- Operon Connector schema is provider-agnostic. The CRM_PROVIDERS array in lib/db/src/schema/crm-connector.ts allows future CRM vendors to be added without a schema rewrite. Operon is provider #1.
- Design Lock is enforced by automated tests. 69 tests guard backend auth, source audit, and UI component audit. Visual regression snapshots are committed and must be updated intentionally.
- Vercel requires two env vars to function in production:
 - VITE_CLERK_PUBLISHABLE_KEY — must be pk_live_ (production key, NOT pk_test_)
 - VITE_CONVEX_URL — production Convex deployment URL
- Git author for all thefsts/* repos: Thefsts

Recommended Next Steps for the New Dashboard Project

- Start with the 10 open Dashboard draft tasks (#54, #58, #59, #66, #75, #76, #79, #80, #81, #82)
- Confirm production Vercel env vars are set correctly (pk_live_ Clerk key + correct CONVEX_URL)
- Verify the Convex deploy key secret is current
- Review docs/product-boundaries.md before planning any new module
- The Design Lock test suite (pnpm run test:design-lock) should pass clean before any new feature work

End of Report — Full Stack Tech Solutions / FSTS Website Operating System